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Here are the 16 multiple-choice questions:

****Q1****

CLASS :

EXAM NO. :

TIME : 3 Hrs

MARKS : 80

What is the main reason why living organisms are classified?

- a) To create a hierarchy of importance
- b) To understand their evolutionary relationships
- c) To make it easier to identify and name them
- d) To study their diversity and characteristics

****Q2****

What is the scientific term for the classification of organisms?

- a) Systematics
- b) Taxonomy
- c) Biology
- d) Evolution

****Q3****

Which of the following is NOT a taxonomic category?

- a) Species
- b) Genus
- c) Order
- d) Color

****Q4****

What do we learn from the identification of individuals and populations?

- a) Their physical characteristics only
- b) Their evolutionary relationships
- c) Only their common features
- d) Nothing, it's just a way to name them

****Q5****

What is the correct scientific name of Mango?

- a) Mangifera Indica
- b) Mangifera indica
- c) Mango sp.
- d) None of the above

****Q6****

Define a taxon. Give some examples of taxa at different hierarchical levels.

- a) A taxon is a group of organisms with common features, and examples include species, genus, family,
- b) A taxon is a single organism, and examples include cat, dog, human, wheat.
- c) A taxon is a group of related species, and examples include mammals, birds, reptiles.

****Q7****

Can you identify the correct sequence of taxonomical categories?

- a) Species Order Phylum Kingdom
- b) Genus Species Order Kingdom
- c) Species Genus Order Phylum

****Q8****

What is the basis for arranging organisms in taxonomic hierarchy?

- a) Their size, color, and shape
- b) Their evolutionary relationships
- c) Their physical characteristics only
- d) Nothing, it's just a way to name them

****Q9****

Why are classification systems changing every now and then?

- a) Because new organisms are being discovered
- b) Because scientists are making mistakes
- c) Because the rules of taxonomy change frequently
- d) None of the above

****Q10****

What is the term for the study of the relationships among organisms?

- a) Biology
- b) Taxonomy
- c) Systematics
- d) Evolution

****Q11****

Why do scientists use a binomial system of nomenclature?

- a) To make it easier to identify and name organisms
- b) To create a hierarchy of importance
- c) To understand their evolutionary relationships
- d) None of the above

****Q12****

What is the correct scientific name for Housefly?

- a) *Musca domestica*
- b) *Musca indica*
- c) *Musca* sp.
- d) None of the above

****Q13****

What is the term for the study of life forms and living processes?

- a) Biology
- b) Taxonomy
- c) Systematics

d) Evolutionary biology

SECTION B (2 marks each)

Q17. Define a taxon. Give one example of a taxon at the level of Class. (Based on content from page 8)

Q18. Explain why taxonomic studies are useful in agriculture and forestry. (Based on content from page 7)

Q19. What is the difference between systematics and taxonomy? (Based on content from page 7 & 8)

Q20. (Internal Choice)

** (a) ** Explain why common names for organisms can be confusing. (Based on content from page 8)

** (b) ** Table 1.1 lists the taxonomic categories for four organisms. In which Kingdom would you classify a mushroom?

Q21. (Internal Choice)

** (a) ** Briefly describe the role of Ernst Mayr in the field of taxonomy. (Based on content from page 8)

** (b) ** As we move up the taxonomic hierarchy (from species to kingdom), what happens to the number of common characteristics?

SECTION C (3 marks each)

Q22. Explain the importance of identifying similarities and dissimilarities among organisms. How does this help in classification?

Q23. Define a taxon. Give an example of a taxon at the level of *Class* and another at the level of *Family*.

Q24. The text mentions that lower the taxonomic rank, the more characteristics members share. Explain this with an example.

Q25. (Choose one)

** (a) ** Describe the difference between *Systematics* and *Taxonomy*, as explained in the passage.

** (b) ** Explain how the study of taxonomic categories contributes to understanding evolutionary relationships between organisms.

Q26. (Choose one)

** (a) ** Explain why it's important to have a standardized system for naming organisms, rather than relying on common names.

** (b) ** What is the significance of the hierarchical arrangement of taxonomic categories (like species, genus, family, etc.)?

Q27. Using Table 1.1 as a reference, identify the *Order* to which both the housefly and mango belong. Write down the full name of the Order.

Q28. The passage states that "Early man deified some of the inanimate matter and some among the animals." Explain the significance of this statement in the context of early classification systems.

STOP.

SECTION D (Case-based, 4 marks each)

Q29. Read the passage below and answer the questions that follow.

"Taxonomic categories are universal, hierarchical, and based on evolutionary relationships. As we move up the hierarchy, the number of common characteristics increases."

** (a) ** Explain why classification becomes more complex as we move higher up in the taxonomic hierarchy. (2 marks)

** (b) ** What is the importance of studying evolutionary relationships while classifying organisms? (2 marks)

Q30. Read the passage below and answer the questions that follow.

"Early systems of classification were based on the 'uses' of various organisms for human needs. While this was practical, it did not reflect evolutionary relationships."

** (a) ** Why were early classification systems based on the 'uses' of organisms considered inadequate? (2 marks)

** (b) ** [Internal Choice]

** (i) ** Explain how the scientific name *Homo sapiens* reflects the hierarchical classification of humans. (2 marks)

** (OR) **

** (ii) ** Describe the importance of the binomial nomenclature system in biological classification. (2 marks)

Answer Key (for teacher reference - not for inclusion in the question paper)

****Q29.****

****(a)**** As we move higher in the taxonomic hierarchy, the number of common characteristics among members

****(b)**** Studying evolutionary relationships helps us understand the origins of organisms and how they have ad

****Q30.****

****(a)**** Early classification systems based on 'uses' didn't reflect the actual relationships between organisms. T

****(b) (i)**** *Homo* represents the genus, which humans share with other closely related apes, indicating a shar

****(OR)****

****(ii)**** The binomial nomenclature system provides a universally accepted and standardized way to name orga

SECTION E (Long-answer, 5 marks each)

****Q31.**** Describe the importance of taxonomic categories in biological classification. Explain, with examples, h

****Q32.**** Explain the difference between 'systematics' and 'taxonomy'. How has the understanding of evolution

****Q33.**** The text discusses the historical development of classification systems, from early systems based on

****(a)**** Explain why early classification systems based on 'uses' were limited. What factors led to the developm

****OR****

****(b)**** Describe the role of binomial nomenclature in providing a universal and unambiguous naming system f

ANSWER KEY (SECTION A)

Answer: c) To make it easier to identify and name them (1 mark)

Answer: b) Taxonomy (1 mark)

Answer: d) Color (1 mark)

Answer: b) Their evolutionary relationships (1 mark)

Answer: b) *Mangifera indica* (1 mark)

Answer: a) A taxon is a group of organisms with common features, and examples include species, genus, family (1 mark)

Answer: a) Species Order Phylum Kingdom (1 mark)

Answer: b) Their evolutionary relationships (1 mark)

Answer: a) Because new organisms are being discovered (1 mark)

Answer: c) Systematics (1 mark)

Answer: a) To make it easier to identify and name organisms (1 mark)

Answer: a) *Musca domestica* (1 mark)

Answer: a) Biology (1 mark)